

SOLDERING MACHINE INNOVATIONS

Target Better Flexibility And Temperature Control

Are you planning to buy a new soldering system, equipment, or robot for printed circuit board assembly in your manufacturing unit? Here is a quick look at recent features—just what you want to know before investing

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The electronics industry is advancing rapidly in technology, and automation is taking the lead in manufacturing spaces. Automating the soldering process is found to be a more efficient alternative to hand soldering, with the latest technologies and soldering accessories in place. However, there are significant applications that still need hand soldering, and hand soldering stations are not going to go out anytime soon. So, you need to consider the best-in-the-class in both automated and hand soldering, before making the right choice to help you save costs, improve quality and productivity, and ensure standard safety practices.

Trendy features in recent launches

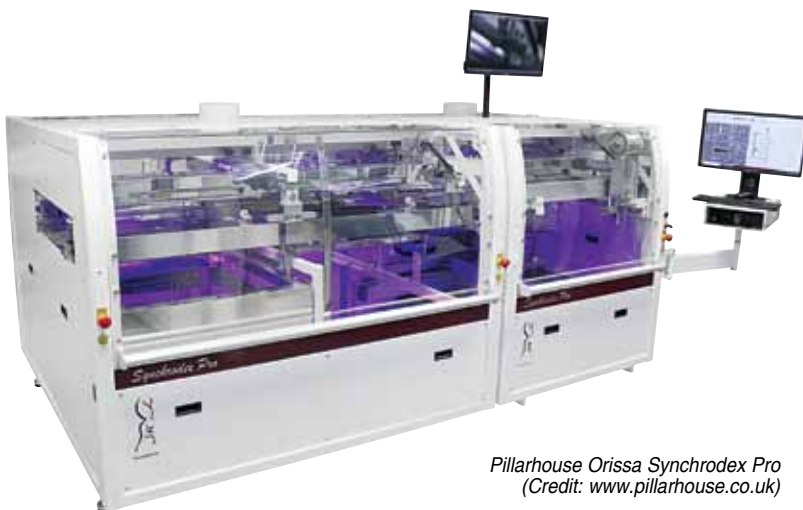
The recent machines and soldering stations have undergone incremental improvements from the previous generation models with more focus to provide manufacturers with ef-

ficient temperature and power control, and motion flexibility.

Capability to handle large PCBs. The industry is handling large printed circuit boards (PCBs) that are used in supercomputers and servers with complex miniature components. Hence the recent launches have more focus on the maximum size of printed circuit boards that can be soldered in the machine.

According to Bittelle Electronics, a PCB assembly house headquartered in Canada, the largest panel size available is 14.50×1.37 metres, which can be fabricated for two-layer boards. The company can fabricate four-layer boards with maximum size of 0.55×1.35 metres. Manufacturers say that there will be additional fabrication cost for PCBs that have sides longer than 50.8 centimetres. For example, Pillarhouse Orissa Synchrodex Pro is a selective soldering system that can process PCBs up to 114.3×60.96 centimetres when using the XL frame size.

Combined systems. The most popular and best soldering equipment comes in integrated packages that are either fully or semi-automatic. Fully automatic are those with soldering modules or soldering robots, safety covers, conveyors, loaders, and unloaders, while semi-automatic are the soldering modules or robots which can be retrofitted into automatic packages to reduce the number of steps and make the process quick and eliminate the need for expensive carriers used in processes like wave soldering.



Pillarhouse Orissa Synchrodex Pro
(Credit: www.pillarhouse.co.uk)